

CO4-AT2-Laboratory Performance

NAME: V Rakshana

SLOT: B

REGNO :192425137

COURSE CODE: CSA0911

QUESTION-1:

CODE:

```
import java.awt.*;
import java.awt.event.*;

public class EventDemo extends Frame implements ActionListener, ItemListener,
WindowListener {

    Button button;

    TextField textField;

    Checkbox checkbox;

    Label result;

    EventDemo() {

        setTitle("Event Source and Type Demo");

        setSize(500, 300);

        setLayout(new FlowLayout());

        button = new Button("Click Me");

        button.addActionListener(this);

        textField = new TextField(20);

        textField.addActionListener(this);

        checkbox = new Checkbox("Select Me");

        checkbox.addItemListener(this);

        result = new Label("Perform an event...");

        add(button);

        add(textField);

        add(checkbox);

        add(result);

        addWindowListener(this);
```

```

        setVisible(true); }

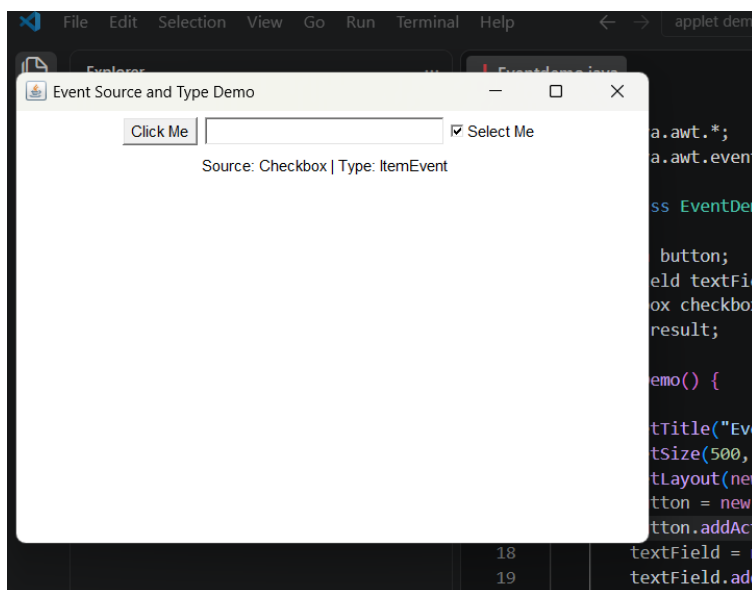
public void actionPerformed(ActionEvent e) {
    if (e.getSource() == button) {
        result.setText("Source: Button | Type: ActionEvent");}
    if (e.getSource() == textField) {
        result.setText("Source: TextField | Type: ActionEvent");}}
public void itemStateChanged(ItemEvent e) {
    result.setText("Source: Checkbox | Type: ItemEvent");}
public void windowClosing(WindowEvent e) {
    System.out.println("Source: Window | Type: WindowEvent");
    dispose();}

public void windowOpened(WindowEvent e) {}
public void windowClosed(WindowEvent e) {}
public void windowIconified(WindowEvent e) {}
public void windowDeiconified(WindowEvent e) {}
public void windowActivated(WindowEvent e) {}
public void windowDeactivated(WindowEvent e) {}

public static void main(String[] args) {
    new EventDemo();}}

```

OUTPUT:



CO4-AT2-Laboratory Performance

NAME: V Rakshana

SLOT: B

REGNO :192425137

COURSE CODE: CSA0911

QUESTION-2:

CODE:

```
import javax.swing.*;
import java.awt.*;
import java.awt.event.*;

public class AppletDemo extends JFrame {
    JButton standaloneButton;
    JButton embeddedButton;
    AppletDemo() {
        setTitle("Java Applet Demonstration");
        setSize(500, 350);
        setLayout(new FlowLayout(FlowLayout.CENTER, 20, 30));
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        JLabel title = new JLabel("JAVA APPLET DEMONSTRATION");
        title.setFont(new Font("Arial", Font.BOLD, 20));
        standaloneButton = new JButton("Stand-alone Applet");
        embeddedButton = new JButton("HTML Embedded Applet");
        standaloneButton.setPreferredSize(new Dimension(180, 40));
        embeddedButton.setPreferredSize(new Dimension(180, 40));
        add(title);
        add(standaloneButton);
        add(embeddedButton);
        standaloneButton.addActionListener(new ActionListener() {
            public void actionPerformed(ActionEvent e) {
                showStandaloneApplet();} });
    }
}
```

```

embeddedButton.addActionListener(new ActionListener() {
    public void actionPerformed(ActionEvent e) {
        showEmbeddedApplet();} });
setLocationRelativeTo(null);
setVisible(true);}

void showStandaloneApplet() {
    JFrame frame = new JFrame("Stand-alone Applet");
    frame.setSize(450, 300);
    frame.setLayout(new FlowLayout());
    JPanel panel = new JPanel();
    panel.setPreferredSize(new Dimension(400, 200));
    panel.setBorder(BorderFactory.createLineBorder(Color.BLUE, 2));
    panel.setLayout(new GridLayout(3, 1));
    JLabel heading = new JLabel(
        "STAND-ALONE APPLET",
        JLabel.CENTER);
    JLabel message = new JLabel(
        "This applet runs independently.",
        JLabel.CENTER);
    JLabel method = new JLabel(
        "It is represented here using Swing.",
        JLabel.CENTER);
    heading.setFont(new Font("Arial", Font.BOLD, 18));
    panel.add(heading);
    panel.add(message);
    panel.add(method);
    frame.add(panel);
    frame.setLocationRelativeTo(null);
    frame.setVisible(true);}

void showEmbeddedApplet() {

```

```
JFrame frame = new JFrame("HTML Embedded Applet");
frame.setSize(550, 400);
frame.setLayout(new BorderLayout());
JPanel htmlPage = new JPanel();
htmlPage.setLayout(new BorderLayout());
JLabel htmlTitle = new JLabel(
    "HTML WEB PAGE",
    JLabel.CENTER);
htmlTitle.setFont(new Font("Arial", Font.BOLD, 20));
JLabel htmlText = new JLabel(
    "The following area represents an embedded applet:",
    JLabel.CENTER);
JPanel appletArea = new JPanel();
appletArea.setLayout(new GridLayout(3, 1));
appletArea.setBorder(
    BorderFactory.createTitledBorder(
        "Embedded Applet Area"));
JLabel appletTitle = new JLabel(
    "HTML EMBEDDED APPLET",
    JLabel.CENTER);
JLabel appletMessage = new JLabel(
    "This applet is embedded inside a webpage.",
    JLabel.CENTER);
JLabel appletInfo = new JLabel(
    "Applet Content",
    JLabel.CENTER);
appletTitle.setFont(
    new Font("Arial", Font.BOLD, 18));
appletArea.add(appletTitle);
appletArea.add(appletMessage);
```

```

appletArea.add(appletInfo);

htmlPage.add(htmlTitle, BorderLayout.NORTH);
htmlPage.add(htmlText, BorderLayout.CENTER);
htmlPage.add(appletArea, BorderLayout.SOUTH);

frame.add(htmlPage);

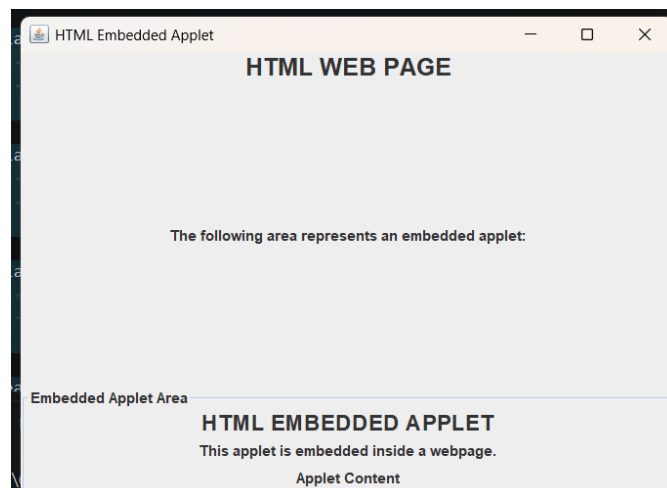
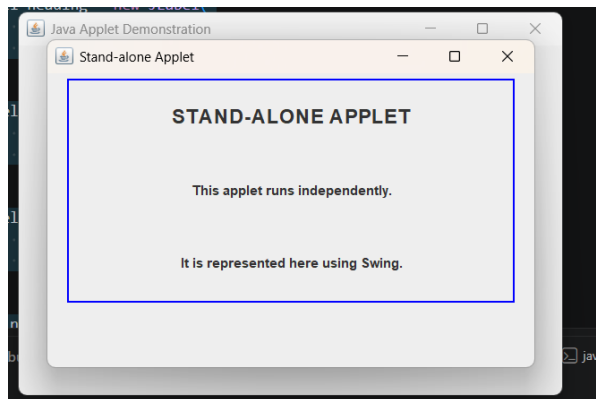
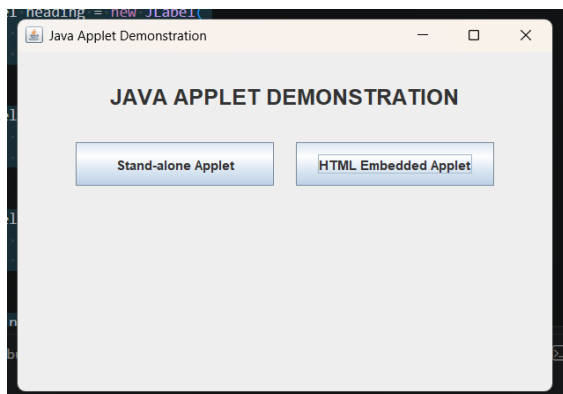
frame.setLocationRelativeTo(null);

frame.setVisible(true);}

public static void main(String[] args) {
    new AppletDemo();
}
}

```

OUTPUT:



CO4-AT2-Laboratory Performance

NAME: V Rakshana

SLOT: B

REGNO :192425137

COURSE CODE: CSA0911

QUESTION-3:

CODE:

```
import java.awt.*;
import java.awt.event.*;

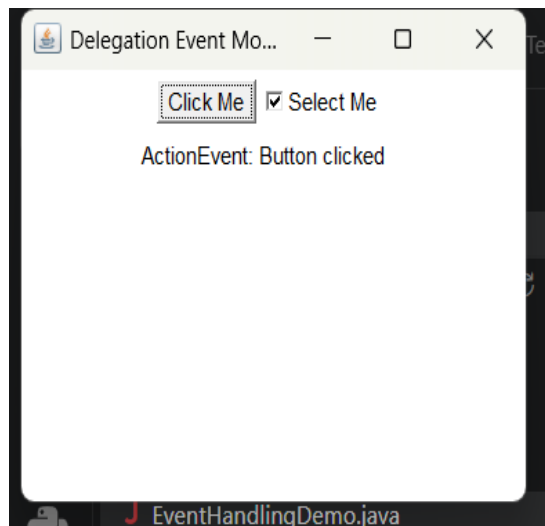
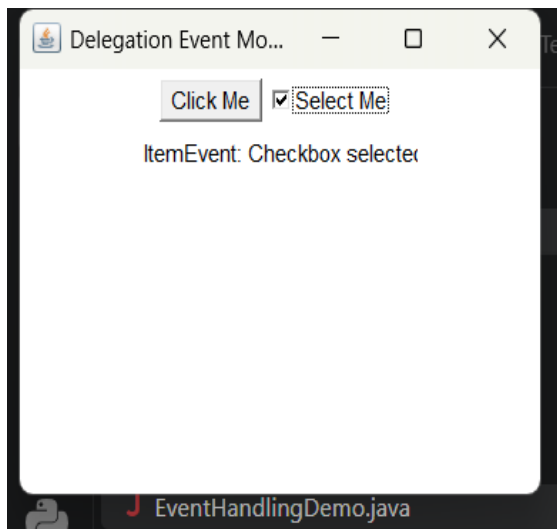
public class EventHandlingDemo extends Frame
    implements ActionListener, ItemListener {
    Button button;
    Checkbox checkbox;
    Label label;

    EventHandlingDemo() {
        setTitle("Delegation Event Model");
        setSize(450, 250);
        setLayout(new FlowLayout());
        button = new Button("Click Me");
        button.addActionListener(this);
        checkbox = new Checkbox("Select Me");
        checkbox.addItemListener(this);
        label = new Label("Perform an event");
        add(button);
        add(checkbox);
        add(label);

        addWindowListener(new WindowAdapter() {
            public void windowClosing(WindowEvent e) {
                dispose();}
        });
        setVisible(true);}
}
```

```
public void actionPerformed(ActionEvent e) {  
    label.setText("ActionEvent: Button clicked");}  
public void itemStateChanged(ItemEvent e) {  
    if (checkbox.getState()) {  
        label.setText("ItemEvent: Checkbox selected");  
    } else {  
        label.setText("ItemEvent: Checkbox deselected");}  
}  
public static void main(String[] args) {  
    new EventHandlingDemo();}}
```

OUTPUT:



CO4-AT2-Laboratory Performance

NAME: V Rakshana

SLOT: B

REGNO :192425137

COURSE CODE: CSA0911

QUESTION-4:

CODE:

```
import javax.swing.*;
import java.awt.*;
import java.awt.event.*;

public class LayoutDemo extends JFrame implements ActionListener {
    JPanel panel;
    JRadioButton flowButton, borderButton, gridButton;
    ButtonGroup group;
    public LayoutDemo() {
        setTitle("Layout Manager Demo");
        setSize(500, 350);
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        setLocationRelativeTo(null);
        panel = new JPanel();
        panel.setLayout(new FlowLayout());
        JButton b1 = new JButton("Button 1");
        JButton b2 = new JButton("Button 2");
        JButton b3 = new JButton("Button 3");
        JButton b4 = new JButton("Button 4");
        JButton b5 = new JButton("Button 5");
        panel.add(b1);
        panel.add(b2);
        panel.add(b3);
        panel.add(b4);
        panel.add(b5);
    }
}
```

```

flowButton = new JRadioButton("FlowLayout", true);
borderButton = new JRadioButton("BorderLayout");
gridButton = new JRadioButton("GridLayout");
group = new ButtonGroup();
group.add(flowButton);
group.add(borderButton);
group.add(gridButton);
flowButton.addActionListener(this);
borderButton.addActionListener(this);
gridButton.addActionListener(this);
JPanel radioPanel = new JPanel();
radioPanel.add(flowButton);
radioPanel.add(borderButton);
radioPanel.add(gridButton);
add(radioPanel, BorderLayout.NORTH);
add(panel, BorderLayout.CENTER);
setVisible(true);}

public void actionPerformed(ActionEvent e) {
    panel.removeAll();
    if (flowButton.isSelected()) {
        panel.setLayout(new FlowLayout());
        panel.add(new JButton("Button 1"));
        panel.add(new JButton("Button 2"));
        panel.add(new JButton("Button 3"));
        panel.add(new JButton("Button 4"));
        panel.add(new JButton("Button 5"));
    }
    else if (borderButton.isSelected()) {
        panel.setLayout(new BorderLayout());
        panel.add(new JButton("North"), BorderLayout.NORTH);
    }
}

```

```

        panel.add(new JButton("South"), BorderLayout.SOUTH);
        panel.add(new JButton("East"), BorderLayout.EAST);
        panel.add(new JButton("West"), BorderLayout.WEST);
        panel.add(new JButton("Center"), BorderLayout.CENTER);
    }
    else if (gridButton.isSelected()) {
        panel.setLayout(new GridLayout(2, 3));
        panel.add(new JButton("Button 1"));
        panel.add(new JButton("Button 2"));
        panel.add(new JButton("Button 3"));
        panel.add(new JButton("Button 4"));
        panel.add(new JButton("Button 5"));
        panel.add(new JButton("Button 6"));
    }
    panel.revalidate();
    panel.repaint();
}

public static void main(String[] args) {
    new LayoutDemo();
}

```

OUTPUT:

